Recent IBD sharing across ancestral groups in Quebec

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Introduction

Identity-By-Descent (IBD) reflects

- Genetic sharing between individuals
- Relatedness
- Demographic history

Objective: Study recent genetic sharing in the diverse population of Quebec

CARTaGENE Data

Quebec-based cohort

N = 29,209 individuals:

- Genetic data (genotyping array)
- Phenotypic data (self-reported)

Diverse origins and ethnic backgrounds

Methods

Step 1: hap-IBD[1] segments detection

Step 2: Louvain clustering[2] + F_{ST}

Step 3: UMAP & HDBSCAN analysis[3]

21 IBD clusters reflecting groups of diverse origins

Geographic origin	Number of clusters	Sample size
Canada	10	23,381
Europe	3	3,392
East-Asia	4	234
Africa	2	560
Latin America	1	525
Mediterranea/North-Africa	1	1,117
Total	21	29,209

Table 1. Identity-by-Descent clusters' origins based on self-reported metadata. Predominent geographic origins were inferred from country of birth.

IBD clustering method vs UMAP & HDBSCAN

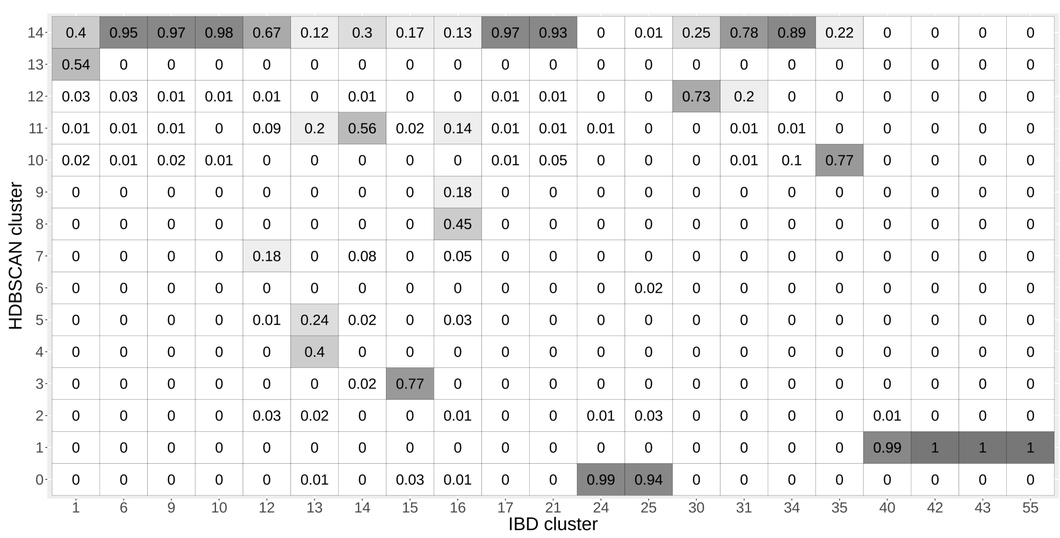

Figure 1. Comparison of IBD clustering and HDBSCAN clustering. Proportion of samples in each IBD cluster that are found in HDBSCAN clusters (rounded to 2 digits).

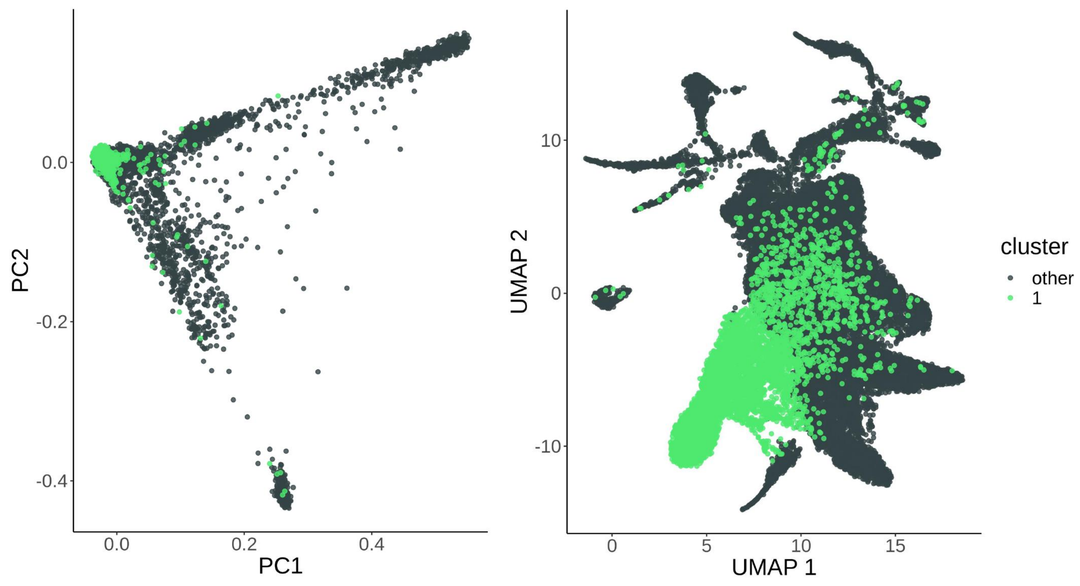

Figure 2. IBD cluster 1 on PCA and UMAP space. Invidivuals in IBD cluster 1 are highlited on: Left. PCA projection against 1000 Genomes Project samples. Right. UMAP on the top 25 PCs of all CARTaGENE samples.

IBD between and within clusters

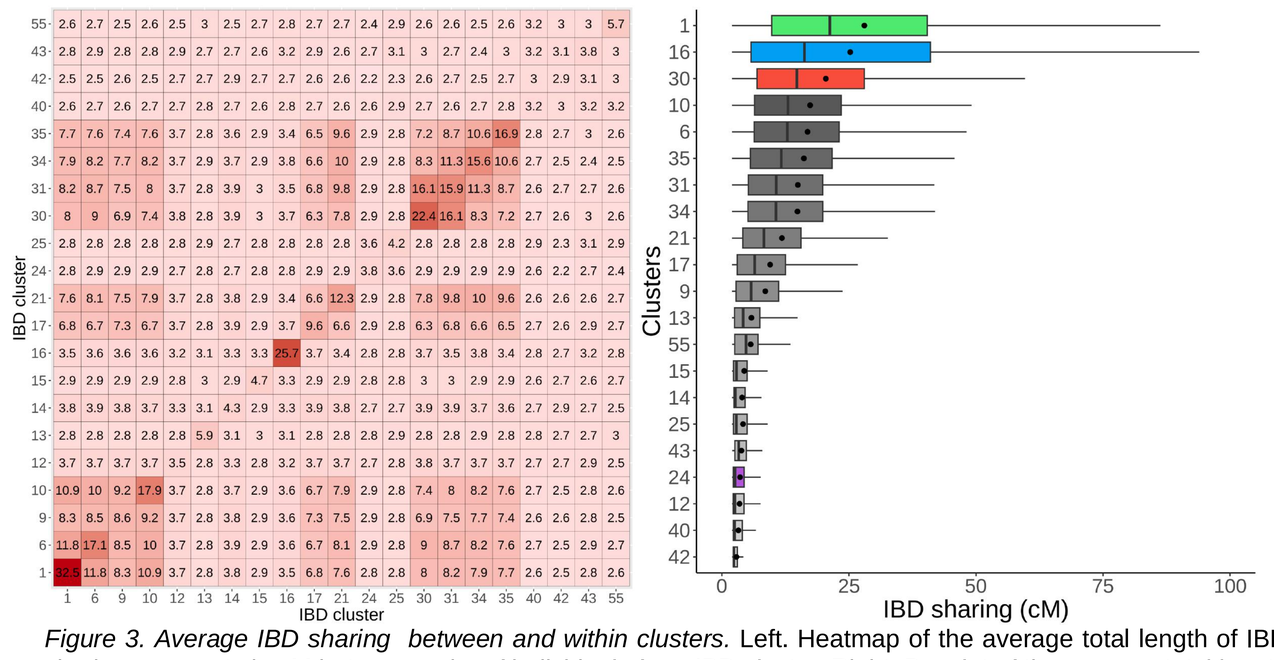

Figure 3. Average IBD sharing between and within clusters. Left. Heatmap of the average total length of IBD sharing segments in cM between pairs of individuals from IBD cluster. Right. Boxplot of the average total length of IBD segment sharing between individuals within each cluster.

Figure 4. IBD cluster labeling and assessment. Cluster 1. Heatmap of recruitment center used to identify and label IBD cluster. *Cluster 16*. Heatmap of self-declared ethnicity used to identify and label IBD cluster. *Cluster 30.* Heatmap of province of birth used to identify and label IBD cluster. *Cluster 24*. Heatmap of places of birth used to identify and label IBD cluster

Conclusions

Description of understudied population subgroups in CARTaGENE
-Diaspora populations-
Description of French-Canadian subgroups with strong IBD sharing
-Founder populations-

Significance

Allow investigation of disease prevalence population subgroups and IBD mapping

Limitations

Self declared Metadata – Limits the inference of cluster origins

References

1. Zhou et al., 2020 . A Fast and Simple Method for Detecting Identity-by-Descent Segments in Large-Scale Data, Am J Hum Genet.
2. Blondel et al., 2008. Fast Unfolding of Communities in Large Networks, J. Stat. Mech.
3. Diaz-Papkovich et al., 2023. Topological stratification of continuous genetic variation in large biobanks, bioRxiv.

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CARTaGENE

McGill UNIVERSITY

GitHub